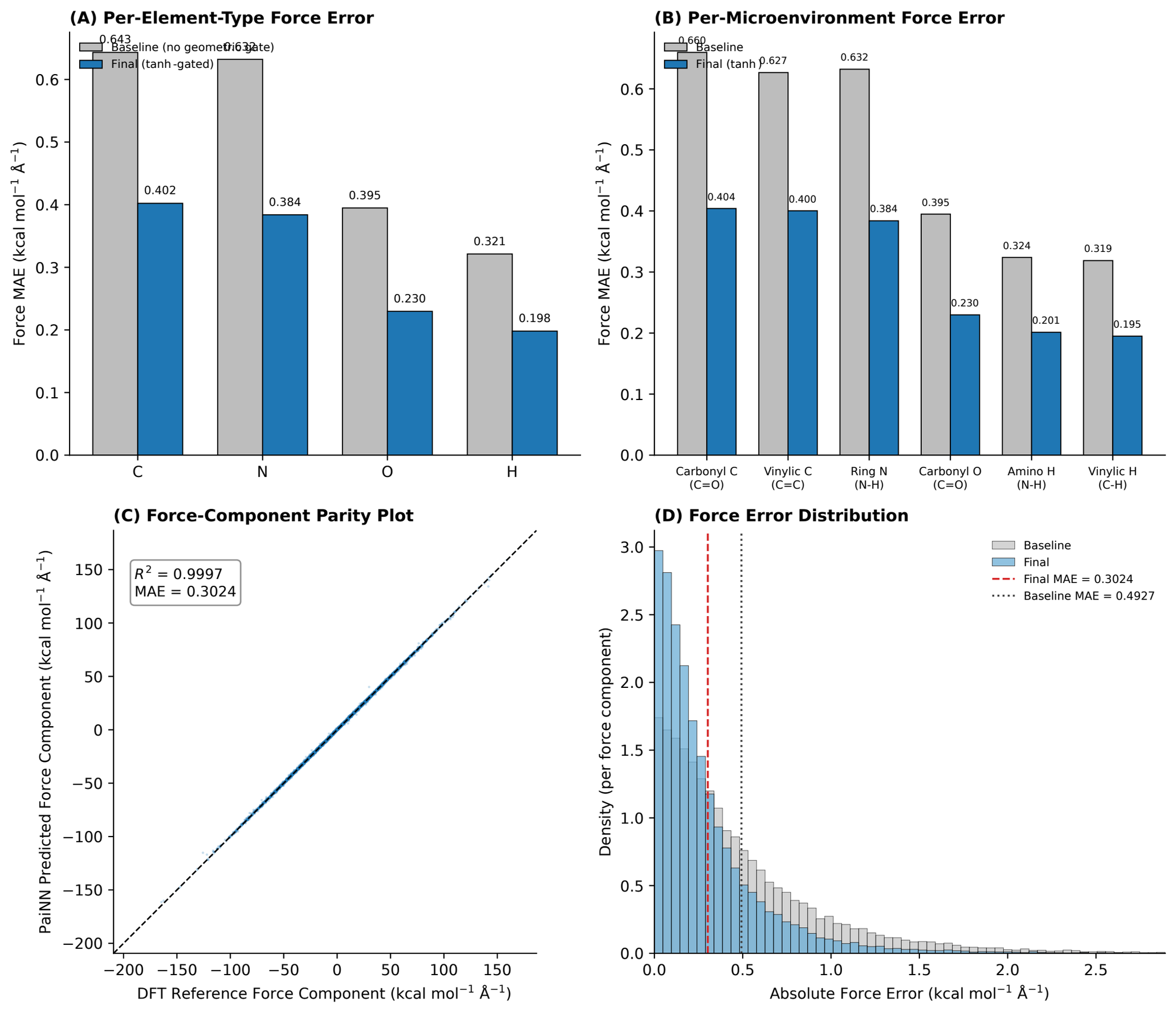

(A) Per-Element-Type Force Error
Baseline (no geometric gate)
Final (tanh-gated)
Force MAE (kcal mol⁻¹ Å⁻¹)
0.643
0.402
0.632
0.384
0.395
0.230
0.321
0.198
C
N
O
H
(B) Per-Microenvironment Force Error
Baseline
Final (tanh)
Force MAE (kcal mol⁻¹ Å⁻¹)
0.660
0.404
0.627
0.400
0.632
0.384
0.395
0.230
0.324
0.201
0.319
0.195
Carbonyl C (C=O)
Vinylic C (C=C)
Ring N (N-H)
Carbonyl O (C=O)
Amino H (N-H)
Vinylic H (C-H)
(C) Force-Component Parity Plot
R^2 = 0.9997
MAE = 0.3024
PaiNN Predicted Force Component (kcal mol⁻¹ Å⁻¹)
DFT Reference Force Component (kcal mol⁻¹ Å⁻¹)
(D) Force Error Distribution
Baseline
Final
Final MAE = 0.3024
Baseline MAE = 0.4927
Density (per force component)
Absolute Force Error (kcal mol⁻¹ Å⁻¹)